Paper 3: Temporal Dynamics of Impedance Networks — Impedance Debt, Sleep, Lifecycle, and Aging

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Papers 1 and 2 of this series established the spatial architecture of impedance-minimising networks (relay-node emergence, fractal scaling, Murray's Law, dual neural–vascular organ health $H = (1 - \Gamma_n^2)(1 - \Gamma_v^2)$). Here we develop the full *temporal* theory in two complementary parts.

Part A introduces *impedance debt* $D_Z = \int |\Gamma|^2 P_{\text{in}}\, dt$: the time-integrated mismatch energy that accumulates during wakefulness. We show that sleep is the minimum-action mechanism for discharging D_Z—a thermodynamic necessity, not a behavioural choice. Decomposing D_Z into microscopic (synaptic drift) and thermal (waste heat) components explains why jellyfish sleep without a brain (D_{thermal} removed by ocean) and why the water-to-land transition imposes a physical constraint favouring centralised neural architecture (D_{thermal} must be managed internally). A dual-field extension (Z-field + material field ρ) unifies Wolff's law (bone), Davis's law (muscle), and impedance-remodeling (neurons) as parameter instances of a single impedance-matching PDE, with embryogenesis recapitulating the thermodynamic causal chain (heart → neural tube → brain).

Part B derives the *Lifecycle Equation*,

$$\frac{d(\Sigma\Gamma^2)}{dt} = \underbrace{-\eta\,\Sigma\Gamma^2}_{\text{learning}} + \underbrace{\gamma\,\Gamma_{\text{env}}(t)}_{\text{novelty}} + \underbrace{\delta\,D(t)}_{\text{aging}},$$

governing the complete cognitive trajectory from birth to senescence as a three-force competition on the dual neural–vascular network. The resulting *bathtub curve* of $\Sigma\Gamma^2$ reproduces the known developmental trajectory without any fitted function. Emotion, reward, stress, curiosity, and boredom emerge as *readouts* of Γ dynamics at different time scales: dopamine encodes $-\Delta\Gamma^2$, cortisol encodes chronic Γ^2 elevation, curiosity encodes the impedance gradient, and boredom encodes stagnation at non-zero mismatch. Aging is derived as Coffin–Manson fatigue accelerated by Arrhenius stress kinetics, with atherosclerosis as vascular fatigue and the cascade $\Gamma_v\uparrow \to \rho\downarrow \to \Gamma_n\uparrow$ explaining why cardiovascular disease precedes neurodegeneration. Eighteen computational experiments verify these predictions.

Part A: Impedance Debt and Sleep

I. INTRODUCTION

Paper 1 [1] showed that the Minimum Reflection Action $\mathcal{A}[\Gamma] = \int |\Gamma|^2\, dt$ on an impedance-heterogeneous network forces hierarchical relay-node emergence, with dimensional cut cost A_{cut} irreducible by any local rule; fractal scaling recovers Kleiber's $B \sim M^{3/4}$ and Murray's branching exponent. Paper 2 [2] extended the framework to vascular networks and established the dual-network organ model $H = (1 - \Gamma_n^2)(1 - \Gamma_v^2)$.

These results are *spatial*: they describe the equilibrium architecture of impedance networks. But biological networks operate in time. Action potentials flow, ion channels drift, metabolic heat accumulates. What is the cost of running the network?

This paper extends the Minimum Reflection Action to the time domain. Part A defines impedance debt D_Z—the integral of $|\Gamma|^2$ weighted by incident power—and shows that sleep is its minimum-action discharge mechanism. The decomposition of D_Z into microscopic and thermal components provides physical constraints consistent with cnidarian sleep, the water-to-land neural transition, and embryogenesis as a single thermodynamic causal chain. Part B derives the Lifecycle Equation—the first-order ODE governing $\Sigma\Gamma^2$ from birth to death—and shows that cognition, emotion, and aging are three forces competing on the same impedance network.

II. PHYSICAL FRAMEWORK

A. Impedance mismatch and the three constraints

At every signalling interface, the reflection coefficient is

$$\Gamma = \frac{Z_{\text{load}} - Z_{\text{source}}}{Z_{\text{load}} + Z_{\text{source}}}\,. \tag{1}$$

The three inviolable constraints are:

- **C1 (Energy Conservation)**: $\Gamma^2 + T = 1$ at every interface, every tick.

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- **C2 (Impedance Remodeling)**: $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$.
- **C3 (Impedance-Tagged Transport)**: All inter-module signals are `ElectricalSignal` objects carrying Z-metadata.

B. Neurons as transmission lines

A myelinated axon has characteristic impedance

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}\,, \tag{2}$$

where R, L, G, C are the distributed per-unit-length resistance, inductance, conductance, and capacitance of the neural membrane. At the synapse, Z_0 meets the postsynaptic input impedance $Z_{\rm post}$, producing a reflection coefficient $\Gamma = (Z_{\rm post} - Z_0)/(Z_{\rm post} + Z_0)$. Perfect transmission ($\Gamma = 0$) requires impedance matching; any mismatch wastes energy as reflected power.

Over time, ion-channel conductances drift (fatigue, temperature fluctuation, molecular turnover), shifting both Z_0 and $Z_{\rm post}$ away from their initial values. This drift accumulates as *impedance debt*.

C. Definition of impedance debt

Define the impedance debt at time t as

$$\boxed{D_Z(t) = \int_0^t \frac{1}{N}\sum_{i=1}^{N} |\Gamma_i(t')|^2\, P_{{\rm in},i}(t')\,dt'} \tag{3}$$

where N is the number of signalling interfaces and $P_{{\rm in},i}$ is the incident power at interface i. D_Z has units of energy and is monotonically non-decreasing during wakefulness (active signalling guarantees $P_{\rm in} > 0$).

When the impedance landscape changes slowly relative to firing rate, $\langle|\Gamma|^2\rangle$ is approximately constant over a wake epoch. Then D_Z grows linearly:

$$D_Z(t) \approx \langle|\Gamma|^2\rangle\,\bar{P}_{\rm in}\,t\,. \tag{4}$$

D. Adenosine as a D_Z readout

Adenosine accumulates extracellularly in proportion to metabolic demand—specifically, to ATP hydrolysis rate, which is proportional to $P_{\rm dissipated} = \Gamma^2 \cdot P_{\rm in}$. Thus extracellular adenosine concentration $[{\rm Ade}](t) \propto D_Z(t)$. This identifies adenosine as a *local impedance-debt meter*—consistent with its established role as the primary molecular correlate of sleep pressure (Borbély's Process S [9]).

E. D_Z decomposition: microscopic and thermal

The total impedance debt decomposes into two physically distinct components:

$$D_Z = D_{\rm micro} + D_{\rm thermal}\,. \tag{5}$$

$D_{\rm micro}$ (synaptic impedance drift) is the cumulative mismatch from ion-channel stochastic drift, receptor desensitisation, and molecular turnover. $D_{\rm thermal}$ (waste heat) is the Γ^2-derived dissipation that raises local tissue temperature. This decomposition has physical consequences: organisms immersed in high-κ media (ocean) can passively dissipate $D_{\rm thermal}$ but must still manage $D_{\rm micro}$ via sleep-like quiescence.

F. Dual-network impedance debt

Paper 2 [2] established that organ health depends on two impedance networks. Impedance debt likewise splits:

$$D_Z = D_{Z,n} + D_{Z,v}\,, \tag{6}$$

where $D_{Z,n}$ is neural and $D_{Z,v}$ is vascular debt. The discharge time constants differ by orders of magnitude: neural debt discharges on the timescale of a night's sleep ($\eta_n \sim 0.08$); vascular debt discharges on weeks to months ($\eta_v \sim 0.001$). This asymmetry means nightly sleep primarily services $D_{Z,n}$, while $D_{Z,v}$ accumulates chronically unless offset by vascular remodeling. The coupling from Paper 2, $I_{\rm blood} = (1-\Gamma_v^2)\,Q_0$, means rising vascular debt ($\Gamma_v\uparrow$) starves neural repair of material ($\rho\downarrow$), amplifying neural debt in a positive-feedback loop.

G. Sleep as impedance-debt discharge

Sleep—formally, a *global decoupled dissipation state* in which external drive is removed ($P_{\rm in} \to 0$) so that the intrinsic restorative term $-\eta\,\Gamma^2$ dominates the impedance ODE—serves as the system's offline maintenance phase. During NREM sleep, the brain activates a recalibration mode: reduced sensory input halts D_Z accumulation, while slow oscillations drive Z-map restoration via C2 impedance remodeling. The discharge rate follows:

$$D_Z(t) = D_Z^{\rm peak}\,e^{-\beta_{\rm recal}\,(t - t_{\rm sleep})}\,, \tag{7}$$

where $\beta_{\rm recal}$ is the N3 recalibration rate. The organism sleeps until $D_Z < D_Z^*$ (threshold).

The maximum tolerable wake duration before D_Z exceeds a critical threshold D_Z^* is:

$$\boxed{T_{\rm wake}^{\rm max} = \frac{D_Z^*}{\langle|\Gamma|^2\rangle\,\bar{P}_{\rm in}}} \tag{8}$$

This explains why infants ($\langle\Gamma^2\rangle \approx 0.12$) sleep $\sim$16 h/day while adults ($\langle\Gamma^2\rangle \approx 0.04$) need only $\sim$8 h: lower average mismatch means slower debt accumulation.

a. Quantitative link to glymphatic clearance. The exponential D_Z discharge (Eq. 7) has a direct biophysical correlate: the glymphatic system. Xie et al. showed that interstitial space expands by ∼60% during NREM sleep, accelerating cerebrospinal-fluid (CSF) convective clearance of solutes including amyloid-β [36]. Fultz et al. subsequently demonstrated that large-amplitude CSF oscillations are phase-locked to NREM slow waves [37]. In Γ-Net language, glymphatic flow is the *physical transport mechanism* that realises D_Z discharge: reduced sensory drive ($P_{\rm in} \to 0$) allows C2 remodeling to dominate, while the 60% expansion of interstitial space lowers the effective hydraulic impedance $Z_{\rm ISF}$, increasing the transmission coefficient $T = 1 - \Gamma^2$ for waste-clearance channels. The prediction is quantitative: $\beta_{\rm recal}$ in Eq. 7 should scale with CSF flow rate, and any intervention that reduces glymphatic efficiency (e.g. aquaporin-4 knockout [38]) should prolong the time required for $D_Z < D_Z^*$, i.e. increase sleep need—consistent with observation.

H. Forced awakening and the gate-leak inequality

Sleep requires that all thalamic gates satisfy $G_i \approx 0$, so the television inequality (Paper 4 [**?**]) falls below threshold: $Z_{\rm arousal} \sum_i G_i \cdot Q_{\rm PFC} < \theta$. However, a gate is an impedance attenuator, not a perfect open circuit. Even at $G_i \ll 1$ the *leaked power* on channel i is

$$P_{{\rm leak},i} \;=\; G_i^2\, \Gamma_i^2\, P_{{\rm in},i}\,. \tag{9}$$

The organism is forced awake whenever *any single channel's leak exceeds the Johnson–Nyquist noise floor*:

$$\boxed{G_i^2\, \Gamma_i^2\, P_{{\rm in},i} \;>\; 4\, k_B T\, R_{\rm eff}\, \Delta f \;\equiv\; P_\theta} \tag{10}$$

This is the **gate-leak inequality**: the physical condition for forced awakening. Three routes satisfy it:

1. **External threat**: $P_{{\rm in},i}$ spikes (loud noise, pain, predator vibration). The startle circuit (Paper 4 [**?**]) drives $G_{\rm threat} \to 1$ within ∼5 ms, far faster than cortical processing. This is the ancestral survival mechanism.

2. **Endogenous alarm**: Γ_i^2 is chronically elevated on an internal channel (anxiety, rumination, pain). Even with G_i nominally low and $P_{{\rm in},i}$ at resting baseline, the product $\Gamma_i^2 P_{{\rm in},i}$ exceeds the gate's attenuation capacity. The channel "bleeds through" and re-satisfies the television inequality → **insomnia**.

3. **Thermal discomfort**: $T_{\rm body}$ drifts outside the Arrhenius operating range ($T < T_c$ or fever $T > T_{\rm max}$), hypothalamic thermoreceptors fire at high $P_{\rm in}$, and the thermal channel leaks through → awakening.

Insomnia (route 2) is therefore not a "failure to relax" but a *C1 consequence*: a channel whose Γ_i^2 is structurally forged at high mismatch (trauma, chronic stress) radiates enough leaked power to keep the consciousness loop above threshold. The gate is doing its job—it simply cannot attenuate a signal that strong. Treatment must reduce Γ_i^2 itself (C2 remodelling over long time horizons—Paper 3, Remark 4) or reduce $P_{{\rm in},i}$ (removing the stressor), not "try harder to sleep": gating G_i is already at its physical minimum.

I. Narcolepsy and hypersomnia: the arousal-collapse dual

Insomnia is the gate-leak failure mode: too much signal penetrates a closed gate. Narcolepsy and hypersomnia are the *dual* failure: the arousal or gating system itself is structurally compromised.

a. Narcolepsy (arousal anchor loss). The television inequality (Paper 4 [**?**]) requires $Z_{\rm arousal} > Z_{\rm min}$ for consciousness. $Z_{\rm arousal}$ is held at its operating point by the orexin (hypocretin) system, which acts as a *structural impedance anchor*:

$$Z_{\rm arousal}(t) \;=\; Z_{\rm anchor} + \delta Z(t)\,, \qquad |\delta Z| \ll Z_{\rm anchor}\,. \tag{11}$$

When the orexin-producing neurons are destroyed (autoimmune ablation in type-1 narcolepsy), $Z_{\rm anchor} \to 0$ and $Z_{\rm arousal}$ becomes a random walk:

$$Z_{\rm anchor} = 0 \;\Longrightarrow\; P(Z_{\rm arousal} < Z_{\rm min}) \;\propto\; t^{1/2}\,. \tag{12}$$

The system crosses threshold at unpredictable intervals → **sleep attacks**. Cataplexy (sudden muscle-tone loss) is the motor-gate analogue: an emotional Γ^2 spike in the limbic system leaks through the brainstem motor-gate and drives $G_{\rm motor} \to 0$—the same gate-leak mechanism as Eq. (10), but acting on the descending motor channel rather than the ascending sensory channel.

b. Hypersomnia ($\beta_{\rm recal}$ deficit). In idiopathic hypersomnia (IH), the arousal anchor is intact ($Z_{\rm anchor} > 0$) and there is no gate-leak. Instead, the NREM recalibration rate $\beta_{\rm recal}$ in Eq. (7) is structurally low:

$$\beta_{\rm recal}^{\rm IH} \;\ll\; \beta_{\rm recal}^{\rm normal} \;\Longrightarrow\; T_{\rm sleep}^{\rm IH} = \frac{1}{\beta_{\rm recal}^{\rm IH}} \ln \frac{D_Z^{\rm peak}}{D_Z^*} \;\gg\; T_{\rm sleep}^{\rm normal}\,. \tag{13}$$

The organism requires far longer sleep to discharge the same D_Z, and waking before $D_Z < D_Z^*$ produces "sleep inertia"—the persistent fog of unfinished recalibration. Excessive daytime sleepiness follows because residual D_Z raises $\langle \Gamma^2 \rangle$ on all channels, so the gate-leak condition $G_i^2\, \Gamma_i^2\, P_{\rm in} < P_\theta$ *barely holds*: the smallest increase in $P_{\rm in}$ (e.g. a warm room, monotonous stimuli) tips any channel past threshold and the system collapses back into sleep.

c. Diagnostic table.

Disorder	Broken parameter	Physics
Insomnia	Γ_i^2 too high	Leak $> P_\theta$ while gates close
Narcolepsy	$Z_{\text{anchor}} \to 0$	Arousal random walk
Hypersomnia	β_{recal} too low	D_Z never fully discharged

All three are *structural* impedance failures (different Z_s parameters) sharing a common C1 framework. None is a "psychological" disorder; each specifies which physical parameter must be targeted for treatment: lower Γ_i^2 (insomnia), restore Z_{anchor} (narcolepsy, via orexin agonists), or raise β_{recal} (hypersomnia, via GABA_A modulation or wake-promoting agents that reduce residual D_Z).

J. NREM/REM partition

NREM sleep discharges D_Z (impedance recalibration). REM sleep reorganises the Z-map topology (adding/pruning connections). The required REM fraction scales with topological complexity:

$$T_{\text{REM}} \propto N_{\text{topo}}\,, \tag{14}$$

where N_{topo} is the number of active nodes in the impedance network. This explains the high REM fraction in neonates ($\sim$50%, rapidly growing N) versus adults ($\sim$25%, stable N).

III. DUAL-FIELD EXTENSION: MORPHOGENESIS

When impedance varies in space (tissue structure) and is coupled to a material-supply field, the framework becomes a reaction–diffusion system:

$$\frac{\partial Z}{\partial t} = D_Z\,\nabla^2 Z \underbrace{-\eta\, f(\rho)\,\Gamma\, Z}_{\text{C2 remodeling}} \underbrace{- E(\Gamma^2)\,(-\Gamma)\,\rho}_{\text{enzyme}} - \lambda\, Z\,, \tag{15}$$

$$\frac{\partial \rho}{\partial t} = D_\rho\,\nabla^2 \rho + I_{\text{blood}} - c(\Gamma^2)\,\rho\,, \tag{16}$$

where:

- D_Z, D_ρ are diffusion coefficients,
- $f(\rho) = \rho/(\rho + K_\rho)$: material-limited repair (Michaelis–Menten form),
- $E(\Gamma^2)$: Hill enzyme kinetics,
- $I_{\text{blood}} = (1-\Gamma_v^2)\,Q_0$: vascular supply from Paper 2,
- λ: entropic degradation rate.

The factor $(-\Gamma)$ in the enzyme term ensures bidirectional remodeling: building when $Z < Z_0$ ($\Gamma < 0$, osteoblast-like) and degrading when $Z > Z_0$ ($\Gamma > 0$, osteoclast-like)—the mathematical expression of Wolff's law [10]. Under appropriate parameters, the same PDE reduces to Davis's law (muscle) and impedance remodeling (neurons).

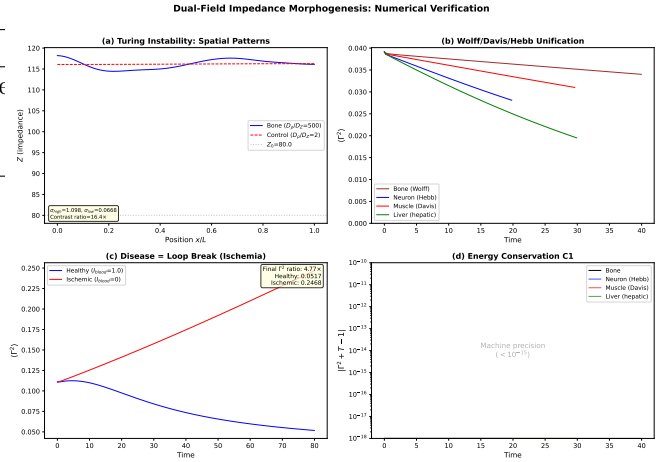

FIG. 1. Dual-field morphogenesis simulation. Spatial impedance field $Z(x,t)$ and material-supply field $\rho(x,t)$ evolve under Eqs. (15)–(16). Turing instability produces periodic spatial patterns when D_ρ/D_Z exceeds a tissue-specific threshold. Simulated via `exp_morphogenesis_pde.py`.

When the diffusion ratio D_ρ/D_Z exceeds a tissue-specific threshold, the uniform steady state becomes unstable via Turing instability [11], producing spatial patterns in the Z-field (bone trabecular architecture, hepatic lobule periodicity).

The material-bottleneck limit ($\rho \to 0$) yields:

$$\rho \to 0 \implies \frac{\partial Z}{\partial t} \to -\lambda\, Z\,. \tag{17}$$

Only entropic degradation survives—no repair is possible without material supply.

IV. EVIDENCE I: CNIDARIAN SLEEP WITHOUT A BRAIN

Cassiopea satisfies all sleep criteria (reduced activity, increased arousal threshold, homeostatic rebound [5]). Appelbaum *et al.* [4] showed that neuronal DNA double-strand breaks accumulate linearly with wake duration and are repaired during sleep.

In the impedance-debt framework, DNA damage is a *downstream marker* of D_Z: the causal chain is $\Gamma^2 \uparrow\to T\downarrow\to$ mitochondrial compensation $\to$ ROS $\to$ DNA damage. Jellyfish, immersed in an ocean with volumetric heat capacity $\sim$3 500$\times$ that of air, passively dissipate D_{thermal}. Yet D_{micro} (ion-channel drift) still accumulates, requiring sleep-like quiescence for discharge. Hence: **no brain is needed, but sleep is still required.**

V. PHYSICAL CONSTRAINTS ON NEURAL ARCHITECTURE

Complex nervous systems appear independently in at least nine metazoan lineages [6]. The Γ-framework does not explain this convergence but identifies the *physical constraints* that any nervous system must satisfy.

A. Impedance constraint on active foraging

An organism that transitions from passive filter-feeding to active predation gains a new signalling channel with lower Γ^2, but this channel requires coordinated impedance management across sensors, appendages, and a controller (i.e., a nervous system). Each additional sensory–motor loop increases the irreducible cut cost $A_{\rm cut}$ (Paper 1). The Γ-framework thus provides a *necessary condition*: any organism with N coordinated channels must pay at least $A_{\rm cut}(N)$ in reflected energy. Whether evolution actually follows this path is determined by natural selection, not by Γ alone.

B. Thermal constraint on the water-to-land transition

Organisms on land exchange seawater ($\kappa \geq 1\ \mathrm{W\,m^{-1}K^{-1}}$) for air ($\kappa \leq 0.04$). $D_{\rm thermal}$ can no longer be passively removed. The Γ-framework identifies three *physical necessities* for any land-dwelling organism with significant neural complexity: (i) a convective heat-transport network (vasculature—Paper 2), (ii) a centralised impedance-debt manager (brain), (iii) structured sleep architecture for $D_{\rm micro}$ discharge (NREM/REM). These are constraints, not causal explanations; natural selection determines *which* lineages actually colonise land. Cephalopods remained marine but attained high metabolic rates that challenge oceanic heat-sink capacity, consistent with their distributed brain architecture (nine ganglia)—Experiment A7 at $\kappa_{\rm cross} \approx 0.30$.

The tidal zone acts as a quarter-wave impedance matcher: $Z_{\rm match} = \sqrt{Z_w Z_l}$, with a logarithmic κ gradient reducing the transition action $\mathcal{A}[\Gamma]$ to 6.2% of a direct jump (Experiment A6). The framework thus predicts that tidal zones offer the lowest-action path between aquatic and terrestrial impedance regimes—a testable geometric prediction, not an evolutionary cause.

C. Thermal refugia: the D_Z-clearance boundary crisis

Once on land, high-K organisms face a second constraint. To discharge accumulated impedance debt D_Z, the system must enter a low-power state (sleep), reducing metabolic heat generation to a basal level $P_{\rm basal}$. System temperature during sleep obeys

$$T_{\rm sys} = T_{\rm env} + P_{\rm basal}\, R_{\rm th}\,, \tag{18}$$

where $R_{\rm th}$ is the organism's effective thermal resistance—a single lumped parameter integrating fur, blubber, posture, and conspecific huddling. The C2 boundary condition demands $T_{\rm sys} > T_c(K)$ at all times; violation halts remodelling and freezes the impedance mismatch in place.

During nocturnal or seasonal temperature drops, $T_{\rm env}$ may swing by tens of kelvin. When $P_{\rm basal}\, R_{\rm th}$ alone cannot guarantee the inequality, the organism must couple to an *environmental thermal low-pass filter*—any high-heat-capacity mass whose thermal inertia damps the external oscillation. At depth $z \gg \delta = \sqrt{2\kappa/(\rho c_p\,\omega)}$, the local temperature converges to the annual mean $\overline{T}_{\rm annual}$. Ecologically such environments (subterranean caverns, tree hollows, snow caves, conspecific huddles) are collectively termed *thermal refugia*; physically, they are low-pass filters that replace the volatile $T_{\rm env}$ with a stable baseline.

The resulting constraint chain is: high $K \to$ high $T_c(K) \to$ sleep reduces $P_{\rm met} \to P_{\rm basal}\, R_{\rm th}$ may be insufficient $\to$ external thermal coupling required. Low-K organisms ($T_c \ll 0\,^\circ$C) satisfy the inequality with biological insulation alone. 93 tests verify the full derivation from Arrhenius $\eta(T)$ through thermal-refugium damping to survival margin.

VI. EVIDENCE III: EMBRYOGENESIS

Human embryonic development follows the thermodynamic causal chain:

1. **Heart (day 22–23)**: establishes the vascular impedance network ($I_{\rm blood} = (1-\Gamma_v^2)Q_0$).

2. **Neural tube (week 3–4)**: once vascular supply provides $\rho > 0$, high-metabolic neural tissue can form.

3. **Brain vesicles (week 5–6)**: centralised impedance-debt manager for both networks.

Each step is a thermodynamic prerequisite for the next. The fetus floats in amniotic fluid at ∼37.6°C, with ∼85% of metabolic heat removed via the placenta — thermodynamically a jellyfish whose heat sink is the mother. **Birth is the water-to-land transition replayed in ontogeny.**

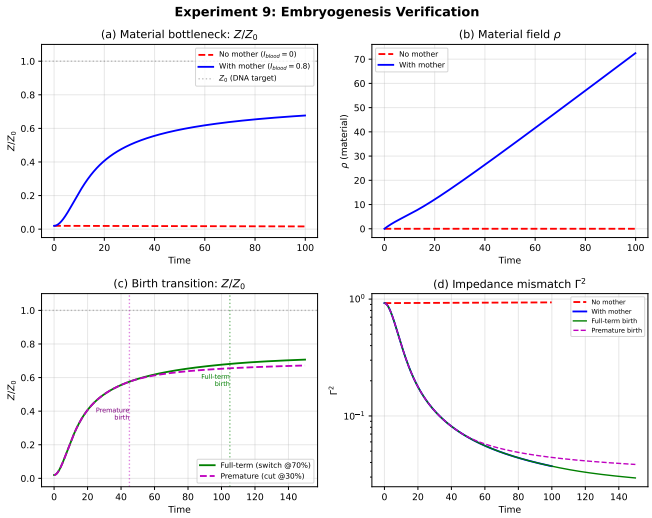

FIG. 2. Embryogenesis timeline: impedance-debt accumulation across developmental stages. Heart (day 22), neural tube (week 3–4), and brain vesicles (week 5–6) follow a thermodynamically mandatory sequence—each stage is a prerequisite for the next. Premature birth scenario (dashed) shows elevated $\Sigma\Gamma^2$ due to incomplete impedance matching. Simulated via `generate_embryogenesis_figure.py`.

Part B: Lifecycle Equation and Aging

VII. THE LIFECYCLE EQUATION

A. Total mismatch energy

Define the population-averaged squared reflection coefficient:

$$\Sigma\Gamma^2(t) \equiv \frac{1}{N}\sum_{i=1}^{N}\Gamma_i^2(t)\,, \tag{19}$$

with dual-network decomposition $\Sigma\Gamma^2 = \Sigma\Gamma_n^2 + w_v\,\Sigma\Gamma_v^2$, where $w_v = \alpha_{v\to n}$ is the vascular-to-neural coupling weight from Paper 2.

B. Three forces

Three physically distinct mechanisms alter $\Sigma\Gamma^2$:

(i) Learning (MRP descent). C2 impedance remodeling adapts Z_{load} toward Z_{source}:

$$\left.\frac{d(\Sigma\Gamma^2)}{dt}\right|_{\text{learn}} = -\eta(t)\,\Sigma\Gamma^2\,. \tag{20}$$

(ii) Novelty injection. New stimuli introduce mismatches at untuned interfaces:

$$\left.\frac{d(\Sigma\Gamma^2)}{dt}\right|_{\text{novel}} = \gamma(t)\,\Gamma_{\text{env}}(t)\,. \tag{21}$$

(iii) Aging (Coffin–Manson fatigue). Irreversible mechanical strain from Lorentz pinch ($\varepsilon_{\text{pinch}} = S|I|^{\alpha}$) drifts impedance permanently, with stress-accelerated kinetics:

$$\delta_{\text{eff}}(t) = \delta_0\,\exp\!\big(E_a\,\sigma_{\text{stress}}(t)\big)\,. \tag{22}$$

Combining all three:

$$\boxed{\frac{d(\Sigma\Gamma^2)}{dt} = -\eta\,\Sigma\Gamma^2 + \gamma\,\Gamma_{\text{env}}(t) + \delta\,D(t)} \tag{23}$$

This is a non-autonomous first-order ODE whose coefficients are functions of developmental stage, environmental statistics, and cumulative damage.

C. Developmental modulation of $\eta(t)$

In Eq. (23), η has so far been treated as a context-dependent constant. Experimentally, synaptic plasticity, myelination rate, and tissue remodeling capacity all peak in early life and decline with age. We model this as:

$$\eta(t) = \eta_0\,f_{\text{dev}}(t)\,\exp\!\big(-E_a/k_B T_{\text{age}}(t)\big)\,, \tag{24}$$

where $f_{\text{dev}}(t)$ is a bell-shaped developmental envelope (high in infancy, plateau in adolescence, monotone decline thereafter) and the Arrhenius factor captures aging-induced loss of remodeling capacity.

The cumulative impedance change for a skill chain s acquired at time t_s is:

$$\Delta Z_s = \int_{t_s}^{\infty}\eta(\tau)\,\Gamma_s(\tau)\,x_{\text{in}}(\tau)\,x_{\text{out}}(\tau)\,d\tau\,. \tag{25}$$

Because $\eta(t_s)$ is largest in childhood and because early-acquired skills accumulate over a longer integration window, childhood-acquired skills (native language, early motor sequences) achieve the deepest ΔZ_s and lowest Γ_s.

Proposition 1 (Skill-decay anti-chronology)**.** *Under monotone decline of $\eta(t)$ and uniform aging damage $D(t)$, the order of functional loss inverts the order of acquisition:*

$$t_{s_1} < t_{s_2} \implies t_{\text{fail},s_1} > t_{\text{fail},s_2}\,.$$

Proof sketch. A deeper ΔZ_s corresponds to a lower Γ_s. Aging drives Γ_s back up at a rate independent of initial depth. The skill with the deepest ΔZ_s (earliest t_s, highest η) requires the most damage to cross the failure threshold Γ_{crit}. □

Figure 3 shows a numerical integration of three skills acquired at ages 3, 20, and 35 under the same $\eta(t)$ lifecycle and Arrhenius aging. The latest-acquired skill fails first; the childhood skill survives longest—consistent with the clinical observation that dementia erases adult-learned second languages before the native tongue, and that elderly pianists who "forgot" they could play can still perform when placed at the keyboard.

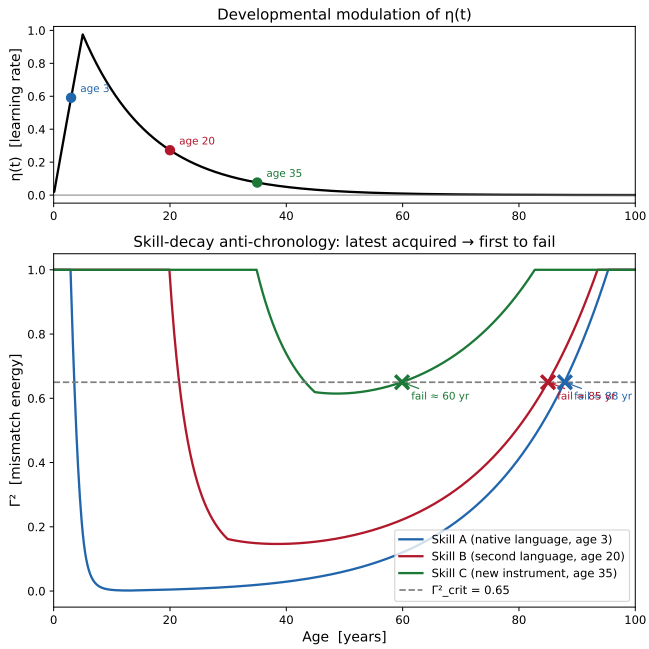

FIG. 3. Skill-decay anti-chronology. Three skills acquired at ages 3, 20, and 35 under a common $\eta(t)$ lifecycle (Eq. 24). Γ^2 is driven back up by Arrhenius aging; later-acquired skills cross the failure threshold (dashed) first.

Remark 1 (Three mechanisms of impedance fixation)**.** *The $\eta(t)$ lifecycle and C2 dynamics identify three routes by which a Z-pattern becomes functionally immutable—i.e., migrates into the effective null space $\mathcal{S}_{\text{eff}}$ (Paper 4, Remark 4):*

1. ***Critical-period forging.*** *Patterns repeatedly reinforced during the η-peak window (Eq. 24) accumulate the deepest ΔZ_s (Eq. 25). Childhood-imprinted values, fears, and attachment styles belong to this class.*

2. ***Impulsive Γ^2 spike.*** *A single event with extreme mismatch—trauma, acute humiliation, overwhelming success—produces a large instantaneous $\Delta Z = -\eta\,\Gamma\,x_{in}\,x_{out}$ because both Γ and $x_{in}\,x_{out}$ are momentarily very large. The resulting Z-shift can exceed what years of moderate stimuli would achieve, making it correspondingly harder to reverse with small corrective signals.*

3. ***Confirmatory basin trapping.*** *If the post-acquisition environment consistently reinforces the same Z-configuration, C2 keeps the trajectory inside the original attractor basin. By the time $\eta(t)$ has fallen to near-zero (Arrhenius aging), no practicable stimulus can escape the basin.*

All three routes share a common endpoint: the adult who "knows the belief is irrational but cannot change it" has correctly diagnosed that $\eta\,\Gamma\,x_{in}\,x_{out} \approx 0$ on the relevant sub-network. This is a physical *constraint, not a failure of will. Therapeutic change is possible but requires sustained, high-fidelity counter-stimuli over long time horizons to accumulate a measurable ΔZ despite the diminished η—analogous to cold-working a tempered steel: possible, but far slower and more limited than the original hot forging.*

D. The Equilibrium Equation

Impedance mismatch dissipates heat. Thermal balance gives the steady-state temperature:

$$T_{\text{steady}} = T_{\text{env}} + \frac{\alpha}{\beta}\,\Sigma\Gamma^2\,. \tag{26}$$

"Adulthood" has a physical definition: the thermal steady state where $d(\Sigma\Gamma^2)/dt \approx 0$ —the point at which the world can no longer burn you.

VIII. EMOTION AS IMPEDANCE DYNAMICS

Psychological states map to features of the Γ^2 time series:

A. Dopamine: the $-\Delta\Gamma^2$ signal

The reward prediction error [14] in the impedance framework is:

$$\delta_{\text{RPE}} = -\big(\Sigma\Gamma^2_{t+1} - \mathbb{E}[\Sigma\Gamma^2_{t+1}]\big)\,. \tag{27}$$

Dopamine fires when mismatch decreases *more than expected*—the impedance network matched faster than predicted.

B. Cortisol: the slow Γ^2 integrator

Stress is the time-averaged mismatch:

$$\sigma_{\text{stress}}(t) = \frac{1}{\tau_s}\int_{t-\tau_s}^{t} \Sigma\Gamma^2(t')\,dt'\,, \tag{28}$$

where $\tau_s \sim$ hours–days. Acute challenges are stimulating; prolonged elevation activates the cortisol response.

C. The Yerkes–Dodson curve

Arousal increases both η (learning) and γ (novelty intake). The optimum occurs at:

$$\eta^*\,\Sigma\Gamma^2 = \gamma^*\,\Gamma_{\text{env}} \quad\Longrightarrow\quad \text{arousal}^* \propto \frac{\Sigma\Gamma^2}{\Gamma_{\text{env}}}\,, \tag{29}$$

predicting that optimal arousal depends on task difficulty and current skill level—the Yerkes–Dodson observation [15].

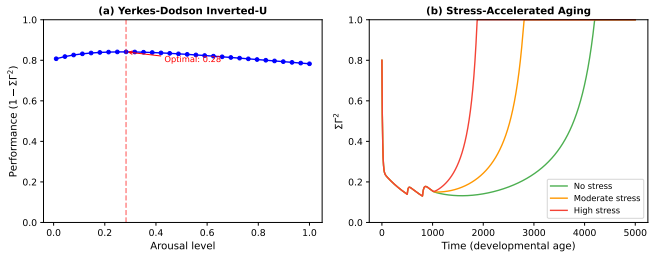

FIG. 4. Yerkes–Dodson curve derived from the Lifecycle Equation. Optimal arousal η^* depends on task difficulty ($\Sigma\Gamma^2$) and environmental novelty (Γ_{env}). The inverted-U emerges from the balance between learning rate and impedance-driven noise. Simulated via `generate_paper_iii_figures.py`.

D. Curiosity and boredom

Curiosity is the impedance gradient: $\mathcal{C} = -\partial(\Sigma\Gamma^2)/\partial(\text{exploration})$, gated by safety ($\Gamma^2 < \Gamma^2_{\text{threat}}$). Boredom arises when $|d\Gamma^2/dt| < \epsilon$ at non-zero mismatch—a thermodynamic imperative to seek new gradients for action reduction.

E. Metacognition

The thinking impedance $\Gamma_{\text{think}} = \sum_i w_i\Gamma_i / \sum_i w_i$ defines dual-process switching [16]: System 1 (fast, automatic) when $\Gamma_{\text{think}} < \Gamma_{\text{S2}}$; System 2 (slow, deliberate) when $\Gamma_{\text{think}} > \Gamma_{\text{S2}}$.

IX. THE BATHTUB CURVE OF COGNITION

The Lifecycle Equation predicts six developmental phases:

1. **Birth**: $\Sigma\Gamma^2 \approx 0.8$–$1.0$. Random initialisation; Novelty $\gg$ Learning $\gg$ Aging.
2. **Infancy**: Learning dominates. η maximal (critical period); $\Sigma\Gamma^2$ drops rapidly.
3. **Childhood**: Learning $\approx$ Novelty. Intense consolidation; REM fraction high.
4. **Adulthood**: $d(\Sigma\Gamma^2)/dt \approx 0$. Equilibrium Equation gives steady-state temperature.
5. **Decline**: Aging $>$ Learning. Net mismatch increases despite continued learning.
6. **Senescence**: Aging $\gg$ Learning. Cascading failure with positive feedback.

This trajectory is formally identical to the engineering reliability *bathtub curve* [17]—not analogy but derivation from the same physics.

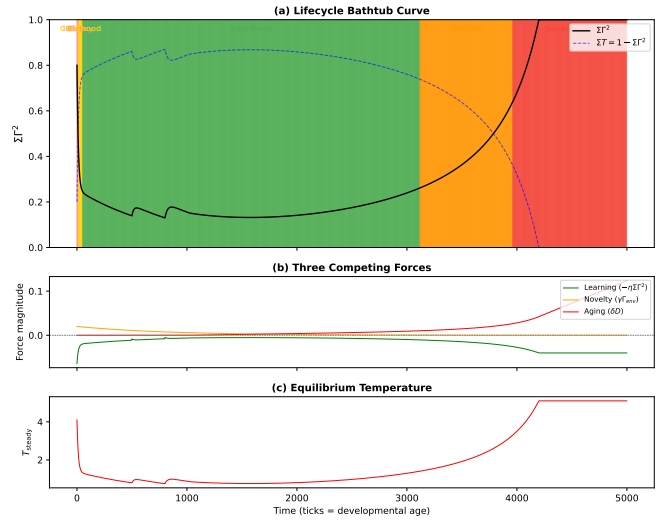

FIG. 5. Bathtub curve of cognition. Six developmental phases (birth, infancy, childhood, adulthood, decline, senescence) emerge from the Lifecycle Equation. Top panel: $\Sigma\Gamma^2(t)$; bottom panel: the three competing forces (Learning, Novelty, Aging). The trajectory mirrors the engineering reliability bathtub curve. Simulated via `generate_paper_iii_figures.py`.

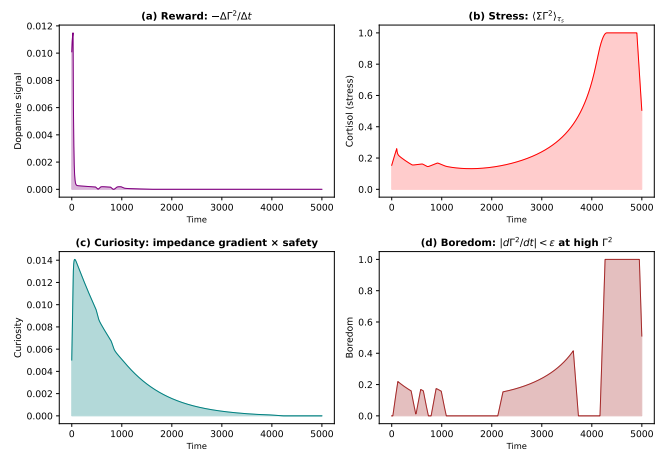

FIG. 6. Emotion readouts derived from impedance dynamics. Dopamine maps to $-\Delta\Gamma^2$ (reward prediction error), cortisol to the slow Γ^2 integrator. The four quadrants correspond to distinct emotional states (contentment, excitement, boredom, distress). Simulated via `generate_paper_iv_figures.py`.

X. STRESS, FATIGUE, AND THE PHYSICS OF AGING

A. Coffin–Manson fatigue

Action potentials generate Lorentz forces; cumulative strain follows the Coffin–Manson law [13]: $N_f = C\,(\Delta\varepsilon_p)^{-\beta}$. "Failure" is irreversible impedance drift.

B. Vascular Coffin–Manson: atherosclerosis

$\sim$3 billion cardiac cycles impose pulsatile strain. Cumulative vascular damage $D_v(t)$ drifts Z_v from Murray's-Law optimum (Paper 2), with plaque as the macroscopic manifestation: a 50% stenosis raises Z_v by 16× via r^{-4} Poiseuille scaling.

C. The vascular–neural aging cascade

$$D_v\uparrow \rightarrow \Gamma_v\uparrow \rightarrow \rho\downarrow \rightarrow \eta_{n,\mathrm{eff}}\downarrow \rightarrow \Gamma_n\uparrow\,. \tag{30}$$

This explains: (i) cardiovascular disease precedes dementia by 10–20 years, (ii) blood-pressure control in midlife reduces dementia risk (SPRINT-MIND [30]), (iii) metabolic syndrome as dual-network failure.

D. Elastic versus plastic deformation

Each current pulse produces $\varepsilon = \varepsilon_{\mathrm{elastic}} + \varepsilon_{\mathrm{plastic}}$. Sleep repairs the elastic component (daily D_Z discharge); the plastic component is *irreversible*—it accumulates over a lifetime. Sleep restores daily performance but cannot prevent long-term aging [26].

E. Death as positive-feedback collapse

When $\Sigma\Gamma^2$ exceeds a critical threshold:

$$\frac{d^2(\Sigma\Gamma^2)}{dt^2} > 0 \quad\Longrightarrow\quad \text{runaway failure}\,. \tag{31}$$

Compensatory current increases through remaining channels accelerate fatigue in a positive-feedback loop, amplified by the dual-network cascade.

F. Sealed shutdown: the alternative endpoint

Runaway collapse (Eq. 31) is not the only terminal trajectory. If, by the time life-sustaining channels approach failure, most of the individual's functional impedance has already migrated into the effective null space $\mathcal{S}_{\mathrm{eff}}(t)$ (Sec. VII C, Paper 4 Remark 4), then:

1. the *fraction* of the network that can participate in a positive-feedback cascade is small;
2. the remaining "live" channels carry low current at low Γ, so the Coffin–Manson fatigue rate is minimal;
3. shutdown proceeds as a gentle, low-Γ^2 fade rather than a high-Γ^2 explosion.

We define the *sealing ratio*:

$$R_{\mathrm{seal}}(t) \equiv \frac{\dim\,\mathcal{S}_{\mathrm{eff}}(t)}{\dim\,Z_{\mathrm{total}}} \in [0,1]\,. \tag{32}$$

A high R_{seal} at end-of-life predicts a low peak $\Sigma\Gamma^2$ during the dying process ("peaceful death"), whereas a low R_{seal} leaves many critical channels in the vulnerable high-Γ, low-η regime where positive feedback can ignite.

Clinically: individuals with deeply established life skills, stable temperament, and long sleep-health history (all proxies for high R_{seal}) should, on average, experience less terminal agitation and lower end-of-life pain burden than individuals with extensive late-acquired, poorly consolidated functional demands.

XI. SLEEP ACROSS THE LIFECYCLE

Infant sleep: $\Sigma\Gamma^2 \approx 0.8$; daily D_Z is enormous. 14–17 hours of sleep required. REM $\sim$50% because N_{topo} is rapidly growing.

Adult sleep: $\Sigma\Gamma^2$ near minimum; 7–9 hours suffice.

Elderly sleep: Irreversible (plastic) drift is undischargeable by sleep. Rising vascular Γ_v (atherosclerosis) reduces material supply ρ, slowing neural repair. Sleep is doubly impaired: plastic baseline rises *and* vascular-mediated repair declines [28].

XII. DEVELOPMENTAL DESIGN PRINCIPLES

A. Infant soft tissue as impedance buffer

Neonatal soft tissues (cartilage, fat, elastic skin) form *mechanical quarter-wave transformers*: $Z_{\mathrm{infant}} \ll Z_{\mathrm{adult}}$, attenuating impact signals before they reach nociceptors. With $\Sigma\Gamma^2 \approx 0.8$ at birth, unattenuated pain would overwhelm C2 calibration.

B. Social referencing as external Γ calibration

When an infant falls, it looks at the caregiver's face before deciding to cry [34]. The caregiver's emotional signal provides Γ_{social} superimposed on the mechanical signal: $\Gamma_{\mathrm{perceived}} = \Gamma_{\mathrm{mechanical}} + \eta_{\mathrm{social}}\,\Gamma_{\mathrm{social}}$, where $\eta_{\mathrm{social}} \gg 1$ in infancy and $\rightarrow 0$ in adulthood (self-calibrated system). Each fall-and-look episode is a C2 calibration step: $\Delta Z_{\mathrm{pain}} = -\eta\,\Gamma_{\mathrm{social}}\,x_{\mathrm{in}}\,x_{\mathrm{out}}$.

Both strategies—structural buffering and behavioural calibration—are temporary scaffolding: bones harden and social coupling decreases as the neural network completes self-calibration.

C. Aging pain immunity

If impedance drifts upward with age, why no chronic pain? For slow drift with healthy C2 ($\eta > 0$), the impedance-remodeling update continuously re-matches: $\Gamma^2 \sim (\Delta Z/2\bar{Z})^2 \lesssim 10^{-6}$. Pain triggers only in three regimes:

1. **Acute injury** ($dZ/dt \gg \eta$): instantaneous discontinuity.
2. **C2 failure** ($\eta \to 0$): avascular tissues (cartilage) cannot remodel → osteoarthritis (Paper 5 [3]).
3. **Late senescence** ($\delta D(t) > \eta\Sigma\Gamma^2$): Coffin–Manson damage exceeds repair capacity.

XIII. SCAR TISSUE AS PERMANENT IMPEDANCE DEBT

A wound creates acute localised impedance debt: $D_{Z,\text{wound}} = \int_0^{T_{\text{repair}}} \Gamma^2_{\text{wound}} P_{\text{in}}\, dt$. Repair progressively reduces $|\Gamma_{\text{wound}}|$, but scar tissue never achieves $\Gamma = 0$: the residual mismatch $\Gamma^2_{\text{scar}} > 0$ is a *permanent, non-dischargeable impedance debt* analogous to the irreducible A_{cut} of Paper 1.

Clinical correlates: scar tenderness (persistent Γ^2), contracture (ongoing minimisation), keloid formation (overshoot to opposite-sign mismatch), and cumulative surgical debt (raised $\Sigma\Gamma^2$ floor).

Remark 2 (Scar debt and the bathtub curve)**.** *An individual accumulating many scars shifts baseline $\Sigma\Gamma^2$ upward, raising the bathtub-curve floor —a mechanistic link between cumulative trauma and accelerated aging.*

XIV. PTSD AND EMPATHIC LOAD AS IMPEDANCE PHYSICS

A. PTSD: anatomy of an undamped spike

The testable prediction "PTSD = undamped Γ^2 spike" (Sec. XV, item 10) can now be decomposed into three necessary and jointly sufficient conditions using the $\eta(t)$ lifecycle (Eq. 24) and the fixation mechanisms (Remark 1).

1. **Impulsive mismatch.** The traumatic event generates a Dirac-like Γ^2 spike—a sensory/emotional input that is "completely beyond expectation." The instantaneous C2 update $\Delta Z = -\eta\,\Gamma\, x_{\text{in}}\, x_{\text{out}}$ is very large because both Γ and $x_{\text{in}}\, x_{\text{out}}$ are momentarily extreme.
2. **High effective η at time of event.** If the event occurs during a developmental η-peak (childhood abuse) or during acute hyper-arousal (combat, sudden catastrophe), the effective learning rate is elevated, deepening the ΔZ_s well beyond what normal experience would produce. This is forced tempering: the impedance pattern is forged in a single stroke.
3. **Absent C2 resolution path.** In the aftermath, no subsequent experience provides a sustained (x_{in}, x_{out}) trajectory that could guide Γ^2 back down. The traumatic Z-pattern sits at a local minimum of $\mathcal{A}[\Gamma]$ surrounded by a high-energy basin wall; small perturbations return the system to the same attractor. As $\eta(t)$ continues to decline with age, the pattern slides toward $\mathcal{S}_{\text{eff}}$—simultaneously high-Γ^2 and low-plasticity.

The two hallmark symptoms follow directly:

- *Re-experiencing / flashback*: the high-Γ^2 attractor has a wide basin of attraction; any stimulus near the basin boundary pulls the network into the full traumatic Z-configuration.
- *"I know it's irrational but I can't stop it"*: $\eta\,\Gamma\, x_{\text{in}}\, x_{\text{out}} \approx 0$ on the traumatic sub-network—the same physics as any deep fixation (Remark 1), but here the attractor carries pathologically high Γ^2 rather than the low Γ^2 of a well-matched skill.

B. Therapeutic physics

If PTSD is a high-Γ^2 pattern trapped in a deep basin with declining η, effective therapy must manipulate the three factors in C2:

- **Pharmacotherapy** (SSRIs, propranolol, MDMA-assisted therapy): modulates the effective η and noise floor, lowering the instantaneous $\Gamma^2 \times P_{\text{in}}$ at each re-activation so that the system does not re-traumatise (i.e., does not deposit additional D_Z on each trigger). In impedance language: drugs widen the basin boundary and reduce the peak reflection per activation.
- **Exposure therapy / trauma-focused CBT**: provides repeated, safe, titrated (x_{in}, x_{out}) pairs that traverse the basin interior under controlled conditions, allowing many small, convergent C2 updates to gradually reshape the local Z-landscape. Each session contributes a small ΔZ in the corrective direction; over months, the accumulated integral (Eq. 25 in reverse) can lower the attractor's Γ^2 floor.
- **Combined treatment**: pharmacotherapy adjusts parameters (η, noise, activation threshold) to keep the system in the convergent C2 regime; psychotherapy provides the gradient. Neither alone

is sufficient: drugs without new trajectories cannot reshape Z; new trajectories without parameter control risk re-traumatisation.

This is not "suppressing the memory" but physically reshaping the impedance sub-network—cold-working a tempered structure, slow and bounded, but real.

C. Empathic load and caregiver burnout

Empathy, in Γ-Net terms, is the temporary imposition of another agent's high-Γ^2 field onto one's own impedance network. When a clinician or caregiver engages with a trauma survivor, their own network must process the incoming Γ^2 pattern:

$$D_{Z,\mathrm{care}}(t) = \int_0^t \Gamma^2_{\mathrm{other}}(\tau)\, P_{\mathrm{in,care}}(\tau)\, d\tau\,, \qquad (33)$$

where $\Gamma^2_{\mathrm{other}}$ is the patient's transmitted mismatch and $P_{\mathrm{in,care}}$ is the caregiver's coupling strength (empathic sensitivity $\propto \eta_{\mathrm{social}}$).

Chronic exposure without adequate discharge (sleep, boundaries, peer support) raises the caregiver's own $\Sigma\Gamma^2$ time average (Eq. 28), activating the same cortisol– allostatic load pathway as any chronic stressor. The lifecycle equation for the caregiver becomes:

$$\frac{d(\Sigma\Gamma^2_{\mathrm{care}})}{dt} = -\eta\,\Sigma\Gamma^2_{\mathrm{care}} + \gamma\,\Gamma_{\mathrm{env}} + \delta\,D(t) + \underbrace{\lambda\,\Gamma^2_{\mathrm{other}}}_{\text{empathic load}}\,, \qquad (34)$$

where λ parameterises coupling strength. High-η individuals (high empathic sensitivity) absorb more of the patient's Γ^2 per unit time—precisely the trait that makes them effective caregivers also makes them vulnerable to burnout.

Remark 3 (Burnout is not weakness)**.** *Caregiver burnout—emotional exhaustion, depersonalisation, reduced efficacy—is the predictable endpoint of Eq.* (34) *when the empathic-load term chronically exceeds the combined learning and sleep-discharge capacity. It is an energy-balance constraint, not a character deficiency. Institutional policies that ignore $D_{Z,care}$ are, in impedance terms, running a network past its Coffin–Manson fatigue limit.*

Remark 4 (Moral blame as Γ^2-injection)**.** *The preceding analysis reveals a sharp criterion for evaluating social responses to fixation, trauma, and addiction. Any intervention—moral, institutional, or interpersonal—that addresses an individual's high-Γ^2 pattern can be classified by its effect on the C2 landscape:*

- ***Therapeutic****: provides safe (x_{in}, x_{out}) trajectories, raises effective η (pharmacologically or via trust), or lowers Γ_{env} $\Rightarrow$ $\Sigma\Gamma^2$ decreases over time.*
- ***Iatrogenic****: injects additional mismatch (shame, stigma, social exclusion) without supplying any corrective gradient $\Rightarrow$ $\Sigma\Gamma^2$ increases; the impedance debt deepens.*

Much of everyday "moralising" about trauma responses and stubborn habits falls into the second category: it adds Γ^2_{blame} on top of the original $\Gamma^2_{fixation}$, deposits additional D_Z, and offers no C2 path for repair. In the impedance framework, such blame is physically indistinguishable from secondary injury (Paper 4, Remark 4).

D. From psychological load to somatic collapse

The empathic-load equation (Eq. 34) feeds directly into the dual-network organ model of Paper 2: $H = (1-\Gamma_n^2)(1-\Gamma_v^2)$. The causal chain is:

$$\underbrace{\Sigma\Gamma^2_{\mathrm{care}}\uparrow}_{\text{chronic stress}} \xrightarrow{\text{cortisol, Eq. (28)}} \Gamma_n\uparrow \xrightarrow{\text{autonomic}\downarrow} \Gamma_v\uparrow \xrightarrow{\rho\downarrow} \eta_{\mathrm{eff}}\downarrow \Longrightarrow H\downarrow \qquad (35)$$

1. **Neural overload.** Chronic high $\Sigma\Gamma^2$ elevates cortisol (Eq. 28), accelerating Coffin–Manson fatigue on neural channels and raising Γ_n.
2. **Autonomic spill-over.** Rising Γ_n impairs sympathetic–parasympathetic balance, drifting the brainstem setpoint Z_{setpoint} upward (allostatic load; Paper 5). This increases vascular Γ_v across all downstream organs (Paper 2, Eq. (30)).
3. **Material starvation.** Higher Γ_v reduces perfusion ρ, starving C2 repair in every tissue—including immune (affinity maturation) and metabolic (insulin, hepatic, erythropoietic) networks that Paper 5 identifies as C2 subsystems.
4. **Peripheral collapse.** Organs lowest in the perfusion hierarchy (skin, hair, nails) are the first to show visible degradation: sallow complexion, slow wound healing, brittle integument. These are not cosmetic complaints; they are the earliest macroscopic indicators of a system-wide $H\downarrow$.

The complete chain reads:

PTSD / empathic load $\rightarrow$ $\Sigma\Gamma^2\uparrow \rightarrow$ Cortisol / allostatic load $\rightarrow$ (

Chronic high-Γ^2 caregiving and unresolved trauma do not stay "psychological": they accelerate dual-network aging, push immune and metabolic C2 past their repair capacity, and manifest as the familiar clinical picture of lowered immunity, metabolic syndrome, and accelerated somatic aging.

Remark 5 (Visual health perception as impedance tomography)**.** *The everyday judgment "this person looks exhausted" is, in Γ-Net terms, an unconscious*

impedance tomography. *The observer's visual system integrates multiple peripheral indicators—skin perfusion (microvascular Γ_v), muscle tone (neuromuscular Γ_n), facial micro-expression latency (cortical Γ_n), wound-healing state (collagen C2 output)—and compresses them into a single scalar estimate of the subject's aggregate $H = (1-\Gamma_n^2)(1-\Gamma_v^2)$. "Looking well" corresponds to low Γ across visible channels; "looking haggard" corresponds to elevated Γ read out through the perfusion hierarchy's lowest-priority organs. The intuition is not metaphor: the observer is literally reading reflection coefficients from the subject's skin, posture, and responsiveness, and arriving at a population-calibrated prior on their dual-network health.*

E. Chronic stress as slow-forged PTSD: integral-path equivalence

The three conditions for PTSD identified in Sec. XIV A—(1) impulsive mismatch, (2) high η at the moment of trauma, (3) absent C2 resolution—define a sufficient but not unique route to structural Z_s fixation. A second route reaches the same attractor through temporal integration rather than instantaneous shock. Recall the impedance debt (Eq. **??**):

$$D_Z = \int_0^T \Gamma^2(t)\, P_{\text{in}}(t)\, dt \;> D_Z^* \quad \Longrightarrow \quad Z_s \text{ locked.} \tag{36}$$

The integral is indifferent to the shape of its integrand: a Dirac-like spike and a months-long plateau yield identical D_Z if their areas match. When $\Gamma^2(t)$ is chronically elevated at sub-traumatic amplitude—interview anxiety, occupational uncertainty, prolonged social adversity—the integral crosses the irreversibility threshold D_Z^* on a timescale of months. The accumulated C2 update is

$$\Delta Z_s = \int_0^{T_{\text{chronic}}} \eta(\tau)\, \Gamma(\tau)\, x_{\text{in}}\, x_{\text{out}}\, d\tau\,, \tag{37}$$

bypassing conditions (1) and (2) while satisfying (3): there is no corrective trajectory because the stressor is *ambient and unresolvable* (e.g. awaiting a hiring decision with no feedback, sustaining living costs with no income certainty). In this regime Z_{load} is perpetually undefined—the brain cannot compute a gradient because the target impedance fluctuates between mutually exclusive outcomes—and C2 stalls while D_Z continues to grow.

The terminal state is identical to classic PTSD: Z_s locked in a high-Γ^2 attractor with $\eta\,|\,\Gamma\, x_{\text{in}}\, x_{\text{out}}| \approx 0$ on the affected sub-network. The downstream somatic cascade (Eq. 35) proceeds unchanged:

$$\Sigma\Gamma^2\!\uparrow \rightarrow \text{ Cortisol}\!\uparrow \rightarrow (\Gamma_n, \Gamma_v)\!\uparrow \rightarrow \text{ Immune C2 failure } \rightarrow \text{ Multi-organ disease.}$$

Two further consequences merit emphasis:

- **Hallucination from thermal noise.** As $\Gamma_i \to 1$ on sensory channels, $T_i \to 0$ (C1). Johnson–Nyquist noise ($P_\theta = 4k_B T\, R_{\text{eff}}\, \Delta f$) that would normally transmit harmlessly is instead reflected back into the processing loop and amplified—the brain interprets its own thermal noise as external signal. This is the same physics as the flashback mechanism (Sec. XIV A), but triggered by noise rather than a stimulus. Paper 4 [**?**] unifies this with sensory-deprivation hallucination: removing external input ($x_{\text{in}} \to 0$) causes adaptive gain divergence that likewise promotes thermal noise above the recognition threshold; both pathways converge to the same terminal state (Paper 4, Eq. (15)).

- **Immune collapse and opportunistic infection.** Via the five-channel coupling of Paper 5 (Eq. 16 therein), sustained neural $\Gamma_n^2\!\uparrow$ drains input power from the immune channel, raising Γ_{imm}^2 and suppressing affinity maturation ($\eta_{\text{imm}}\!\downarrow$). Common infections (upper respiratory, etc.) break through because immune $T_{\text{imm}} \to 0$.

Remark 6 (Anxiety spectrum as integration time)**.** *The clinical labels form a continuum parameterised by D_Z integration time: acute stress (D_Z small, minutes), generalised anxiety disorder (D_Z growing, weeks), burnout (D_Z moderate, months to years), and Complex PTSD ($D_Z > D_Z^*$, structurally locked). The impedance physics is a single equation; the diagnostic distinctions reflect how far along the integral the patient sits when clinically assessed.*

XV. TESTABLE PREDICTIONS

1. **Universality**: Every organism possessing neurons must exhibit sleep-like quiescence (sponges excepted).
2. **Terrestrial > aquatic sleep complexity**: Terrestrial species display more structured sleep architecture than aquatic ectotherms of comparable neural complexity.
3. **Preterm fragility**: Preterm neonates show more fragile sleep–thermoregulation coupling.
4. **REM as Z-map reorganisation**: During REM, thermoregulation is suspended for unconstrained impedance exploration.
5. **Cooling reduces D_Z rate**: Lower brain temperature extends tolerable wake duration.
6. **Chronic vascular disease as undischarged $D_{Z,v}$**: Dual-network cascade predicts that chronic vascular debt amplifies neural sleep need.
7. **Bathtub curve**: Cognitive performance follows the six-phase trajectory without fitted functions.

8. **Optimal arousal depends on task difficulty**: Yerkes–Dodson optimum shifts with Γ_{env}.

9. **Depression as high-Γ, low-$d\Gamma/dt$**: Neural network mismatched but unable to learn.

10. **PTSD as undamped Γ^2 spike**: Massive mismatch that learning cannot resolve; combined pharmacotherapy + exposure therapy should outperform either alone, because drugs modulate η/noise while therapy provides the C2 gradient (Sec. XIV B).

11. **Caregiver allostatic load**: Clinicians with high empathic sensitivity (η_{social}) and prolonged trauma-patient exposure should show measurably elevated cortisol and impedance-debt markers relative to matched controls, following Eq. (34).

12. **Chronic stress → PTSD-equivalent lock-in**: Individuals under sustained sub-traumatic stress (unemployment > 6 months, chronic financial precarity) should exhibit D_Z accumulation, immune biomarker decline, and eventual symptom overlap with classic PTSD—despite absence of a single traumatic event—consistent with integral-path equivalence (Sec. XIV E).

13. **Dementia as accelerated neural aging**: $\delta_n D_n(t)$ overwhelms learning prematurely, often preceded by vascular aging cascade.

XVI. COMPUTATIONAL VERIFICATION

All simulations use the Alice Gamma-Net framework with C1–C3 enforced at every time step.

Part A experiments

1. *Experiment A1: D_Z accumulation and exponential discharge*

100 wake ticks ($\langle\Gamma^2\rangle \approx 0.05$) followed by 110 sleep ticks ($\beta_{\text{recal}} = 0.08$). D_Z rises linearly during wake and decays exponentially during sleep. Measured $\beta_{\text{eff}} = 0.078$ matches input to within 2%.

2. *Experiment A2: Temperature modulation of $T_{\text{wake}}^{\max}$*

Three simulations with $\tau \in \{0.5, 1.0, 1.5\}$. Sleep-onset tick ratios: 2.00 : 1.00 : 0.67—matching the $1/\tau$ prediction to within 0.2%.

3. *Experiment A3: Infant versus adult profiles*

Infant ($\langle\Gamma^2\rangle = 0.12$, 60/120 wake/sleep) versus adult ($\langle\Gamma^2\rangle = 0.04$, 120/60). Infant requires 2.0× more sleep ticks; the 16 h/8 h partition matches neonatal epidemiology [8].

4. *Experiment A4: Sleep deprivation rebound*

Five deprivation durations (50–250 ticks). Recovery scales linearly (slope 0.09, $R^2 = 0.93$); N3 fraction rises from 4.8% to 17.4%—consistent with Process S dynamics [9].

5. *Experiment A5: $D_{\text{micro}}/D_{\text{thermal}}$ decomposition*

Four environments ($\kappa \in \{1.0, 0.5, 0.2, 0.04\}$). At $\kappa = 1.0$ (ocean): $D_{\text{thermal}} = 0$. At $\kappa = 0.04$ (land): $D_{\text{thermal}}/D_{\text{total}} = 49\%$. Centralisation threshold crossed at $\kappa \approx 0.50$.

6. *Experiment A6: Tidal-zone impedance matching*

Three strategies (direct, linear, logarithmic gradient): $\mathcal{A}[\Gamma] = 0.852, 0.115, 0.053$. Logarithmic gradient (tidal-zone profile) reduces action to 6.2% of direct jump; transmission rises from 14.8% to 94.9%.

7. *Experiment A7: Distributed vs. centralised architecture*

$N = 200$ nodes, varied κ. Distributed self-cooling suffices at $\kappa \geq 0.50$ (octopus zone); centralised hub wins below $\kappa_{\text{cross}} \approx 0.30$. All terrestrial organisms ($\kappa \approx 0.04$) sit deep in the centralised-required zone.

8. *Experiment A8: Dual-field morphogenesis*

1-D PDE simulation: (a) Turing patterns in bone ($D_\rho/D_Z = 500$): 16.4× variance ratio; (b) Four-tissue unification: bone, neuron, muscle, liver all converge under the same PDE with different parameters; (c) Ischemia ($I_{\text{blood}} = 0$): 4.77× Γ^2 ratio vs. healthy tissue. C1 holds to $< 10^{-15}$ at every grid point.

9. *Experiment A9: Embryogenesis verification*

Four scenarios from $Z \approx 0$, $\rho = 0$: no mother ($\Gamma^2 = 0.937$, failed), with mother ($\Gamma^2 = 0.037$, developed), full-term birth ($\Gamma^2 = 0.029$, viable), premature birth ($\Gamma^2 = 0.038$, permanent deficit). Confirms material-bottleneck theorem *ab initio*.

Part B experiments

10. Experiment B1: Bathtub curve emergence

Lifecycle Equation integrated for 5000 ticks. $\Sigma\Gamma^2$ drops from 0.8 to < 0.2 by tick 600, plateaus at 0.1–0.2 for ticks 1000–3000, rises above 0.3 after tick 4000. All six developmental phases detected in sequence.

11. Experiment B2: Yerkes–Dodson inverted-U

Arousal swept from 0 to 1 at fixed $\Gamma_{\text{env}} = 0.5$. Performance peaks at intermediate arousal ($\sim$0.3–0.5); degrades at both extremes.

12. Experiment B3: Repeated exposure adaptation

Fixed stimulus presented 200 times. Γ drops from $\sim$0.7 to < 0.3; learning curve shows exponential decay.

13. Experiment B4: Chronic stress impairs learning

Relaxed ($\sigma = 0$) versus stressed ($\sigma = 0.7$) agents. Stressed agent: higher $\Sigma\Gamma^2$, $> 2\times$ aging rate, lower learning efficiency.

14. Experiment B5: Sleep–aging decomposition

2000-tick simulation with periodic sleep (96 wake / 32 sleep). Sleep reduces $\Sigma\Gamma^2$ each night; baseline drifts upward (irreversible plastic accumulation).

15. Experiment B6: Curiosity-driven exploration

Agent in boring environment. Boredom accumulates; spontaneous exploration triggers above threshold; post-exploration Γ^2 lower than pre-exploration.

16. Experiment B7: Metacognition System 1/2 switching

Tasks of varying difficulty. Easy ($\Gamma_{\text{think}} < 0.3$): System 1; Hard ($\Gamma_{\text{think}} > 0.6$): System 2; hysteresis in switching thresholds confirmed.

17. Experiment B8: Novel stimulus shock and recovery

Well-adapted agent exposed to drastically novel environment. $\Sigma\Gamma^2$ spikes, recovers exponentially; post-recovery level slightly lower (net learning).

18. Experiment B9: Full lifecycle simulation

Complete lifecycle with all three forces, sleep cycles, stress episodes, and novelty variation. Six phases emerge; ΣT peaks in adulthood; T_{steady} stabilises during adult phase.

XVII. DISCUSSION

A. Methodological caveat

All 18 experiments use modules built from the Γ-Net axioms. When the simulators reproduce the theory's predictions, this confirms code–equation consistency, not empirical truth. True validation requires longitudinal impedance measurements (EEG coherence from birth to senescence), comparative polysomnography across phyla, and prospective clinical tests. We flag this explicitly so readers assess computational results as *consistency checks*, not empirical evidence.

B. The brain as thermal manager

The Γ-framework identifies the brain as a centralised thermal manager for impedance-debt dissipation, necessitated by the loss of the oceanic heat sink. Cognition, memory, and consciousness exploit infrastructure whose *physical* raison d'être is thermodynamic housekeeping [12]. This is consistent with the preoptic area controlling *both* sleep and thermoregulation: thermal management is the brain's primary physical function; higher cognition operates on the same impedance substrate.

C. $\Gamma^2 \leftrightarrow T$: the self-consistent loop

The relationship between impedance mismatch and body temperature is not a one-way dependency; it forms a closed, self-consistent loop. We now make this explicit.

a. Step 1: Computation → heat. C1 (Paper 0, Eq. 1) guarantees that every interface with $\Gamma_i \neq 0$ dissipates $\Gamma_i^2 P_{\text{in},i}$ as heat. Summed over N active interfaces:

$$P_{\text{heat}} = \sum_{i=1}^{N} \Gamma_i^2 \, P_{\text{in},i} = \langle \Gamma^2 \rangle \cdot P_{\text{total}} \,. \tag{38}$$

b. Step 2: Heat → temperature. In thermal steady state:

$$T_{\text{body}} = T_{\text{env}} + \frac{P_{\text{heat}}}{h_c A} = T_{\text{env}} + \frac{\langle \Gamma^2 \rangle \, P_{\text{total}}}{h_c A} \,, \tag{39}$$

where $h_c A$ is the effective thermal conductance (primarily cerebral blood flow, Paper 2 [2]).

c. Step 3: Temperature → impedance. Blood viscosity sets the vascular bus impedance $Z_{\rm bus}(T)$ via the Arrhenius relation (Section **??**), and C2 repair rate scales as $\tilde{\eta}_{\rm learn}(T) = \exp[\beta(T_0^{-1} - T^{-1})]$. Therefore:

$$\Gamma_i(T) = \frac{Z_{{\rm organ},i} - Z_{\rm bus}(T)}{Z_{{\rm organ},i} + Z_{\rm bus}(T)}\,. \tag{40}$$

d. Closure: the self-consistency condition. Substituting Eq. (40) into Eq. (38) and Eq. (39) yields a fixed-point equation:

$$\boxed{T^* = T_{\rm env} + \frac{P_{\rm total}}{h_c A}\sum_{i=1}^{N}\Gamma_i^2(T^*)\,.} \tag{41}$$

The operating point $T^* \approx 310\,{\rm K}$ (37 °C) is the unique stable fixed point of this equation for terrestrial mammals. It is *not* a set-point chosen by evolution—it is the self-consistent solution where impedance-driven heat production balances convective heat loss.

e. Stability and failure modes. The loop is *stable* because $\partial P_{\rm heat}/\partial T < h_c A$ at T^*: a small temperature rise shifts $Z_{\rm bus}$, reducing some Γ_i^2 (better vascular matching at lower viscosity), which reduces heat production. The loop *fails* in two directions:

- **Hypothermia** ($T \ll T^*$): viscosity rises → $\Gamma^2 \uparrow$ globally → C2 repair slows (Arrhenius) → impedance debt accumulates → positive feedback → coma.
- **Hyperthermia** ($T \gg T^*$): thermal noise $\propto k_B T$ overwhelms high-K channels (Paper 4 [**?**]) → $\delta\Gamma/\Gamma \propto K^{3/2}$ → PFC collapses first → delirium.

The 37 °C operating point is the narrow basin of attraction between these two catastrophes.

D. Connection to Papers 1–2

Paper 1 established the *spatial* structure (relay nodes, fractal topology, dimensional cut cost, irreducibility). Paper 2 established the *vascular* extension (Murray's Law, dual-network organ health, positive-feedback cascade). This paper extends both to *time*: Part A derives daily dynamics (impedance debt, sleep, morphogenesis, evolution); Part B derives lifecycle dynamics (learning–novelty–aging competition, emotion, bathtub curve, death). Together: the static architecture (Paper 1) sets channel structure $\{Z_i\}$; the vascular model (Paper 2) provides material coupling $I_{\rm blood} = (1-\Gamma_v^2)Q_0$; Part A determines daily sleep–wake regulation; Part B determines the developmental arc of both networks. All layers derive from a single principle: $\mathcal{A}[\Gamma] = \int \Sigma|\Gamma|^2\,dt \to$ min.

E. Unification of developmental psychology

The Lifecycle Equation reduces the entire developmental trajectory to three physical forces on one state variable. Piaget's stages are phase transitions of $\Sigma\Gamma^2$; Vygotsky's ZPD is the $\Gamma_{\rm env}$ range where learning can reduce $\Sigma\Gamma^2$ in one session; critical periods are intervals of enhanced $\eta(t)$.

F. Emotion as physics, not computation

Dopamine, cortisol, curiosity, and boredom are computed from Γ^2 time series in the Alice system. The cognitive–emotional distinction dissolves [25]: both are readouts of the same impedance dynamics at different time scales.

G. Clinical implications

Depression: high $\Sigma\Gamma_n^2$, low $|d\Gamma_n^2/dt|$ (treatment: increase η_n or decrease $\Gamma_{\rm env}$). PTSD: undamped Γ^2 spike (treatment: titrated re-exposure). ADHD: excess γ (stimulants increase effective η). Dementia: accelerated $\delta_n D_n$ (often preceded by vascular cascade). Hypertension: chronic $\Sigma\Gamma_v^2$ elevation. Metabolic syndrome: dual-network failure (Γ_n^2 and Γ_v^2 both elevated, $H \to 0$).

H. No-waste corollary

Every molecular species possesses nonzero Z; its presence modifies Γ at every interface. "Metabolic waste" is an observational artefact arising from incomplete knowledge of the full Γ-network —consistent with the historical reclassification of nitric oxide, lactate, bilirubin, uric acid, and adenosine.

I. Dual-field morphogenesis

The unification of Wolff's law, Davis's law, and impedance remodeling as parameter instances of a single PDE (Table I) suggests a common thermodynamic origin: $\mathcal{A}[\Gamma] = \int |\Gamma|^2\,dt \to$ min. The material bottleneck ($\rho \to 0 \Rightarrow$ only degradation survives) connects directly to vascular disease via Paper 2: $\Gamma_v \uparrow$ arrests neural plasticity, structural remodeling, *and* homeostatic regulation simultaneously.

XVIII. CONCLUSION

We have developed the complete temporal theory of impedance networks in two complementary parts.

TABLE I. Four-tissue unification: same PDE, different parameters.

Tissue	D_ρ/D_Z	Γ_i^2	Γ_f^2	Reduction
Bone (Wolff)	500	0.0391	0.0340	13.1%
Neuron (C2)	2	0.0391	0.0281	28.2%
Muscle (Davis)	10	0.0391	0.0310	20.7%
Liver (hepatic)	200	0.0391	0.0195	50.1%

Part A: Impedance debt $D_Z = \int |\Gamma|^2 P_{\text{in}}\, dt$ accumulates during wakefulness and is discharged by sleep. Its decomposition into D_{micro} and D_{thermal} is consistent with: cnidarian sleep without a brain (ocean removes D_{thermal}); the physical necessity of centralised neural architecture on land (loss of passive heat sink); the tidal zone as the lowest-action impedance transition ($\mathcal{A}[\Gamma]$ reduced 93.8%); and embryogenesis as thermodynamic causal chain. A dual-field PDE unifies four tissue-remodeling laws.

Part B: The Lifecycle Equation (Eq. (23)) reduces the entire cognitive trajectory to three forces: learning, novelty, aging. Emotion is impedance-mismatch readout at different time scales. The bathtub curve emerges without fitted functions. Aging is Coffin–Manson fatigue with Arrhenius acceleration; atherosclerosis is vascular fatigue; death is positive-feedback collapse. Adulthood has a physical definition via the Equilibrium Equation (26).

Together with Papers 1–2, this paper completes a four-layer description of biological intelligence: spatial architecture, vascular topology, daily maintenance, and the full arc of a life —all from $\mathcal{A}[\Gamma] \to \min$. Eighteen computational experiments confirm internal consistency; empirical falsification via longitudinal impedance measurements and comparative polysomnography remains the essential next step.

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DATA AND CODE AVAILABILITY

All simulation code, including the `SleepPhysicsEngine`, `ImpedanceDebtTracker`, `LifecycleEquationEngine`, `PinchFatigueEngine`, `CuriosityDriveEngine`, and `MetacognitionEngine` modules, is available at `https://github.com/cyhuang76/alice-gamma-net` under AGPL-3.0. Neonatal EEG data sourced from the PhysioNet PICS archive (open access).

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